

ESTROGEN

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- **Source of Secretion :—**

- **In Non-Pregnant Woman** : Secreted by the **theca interna cells** and **granulosa cells** of ovarian follicles.
 - Secretion is predominant in the **follicular phase** before ovulation.
 - Estrogen is synthesized from androgens, particularly **androstenedione** produced by theca interna cells:



- **In Pregnant Woman** : Secreted by the **placenta**.
- **Chemistry** : C₁₈ steroid hormone.
- **Different Forms in Plasma :—**
 - a. **β-Estradiol** (*Major and most potent form*)
 - b. **Estrone**
 - c. **Estriol** (*Weak / inactive form*)
- **Plasma Levels :—**
 - **In Females (Follicular Phase)** : 30 – 200 pg/mL
 - **In Adult Males** : 12 – 34 pg/mL
- **Half-Life** : 30 – 60 minutes.

BIOSYNTHESIS, TRANSPORT & METABOLISM

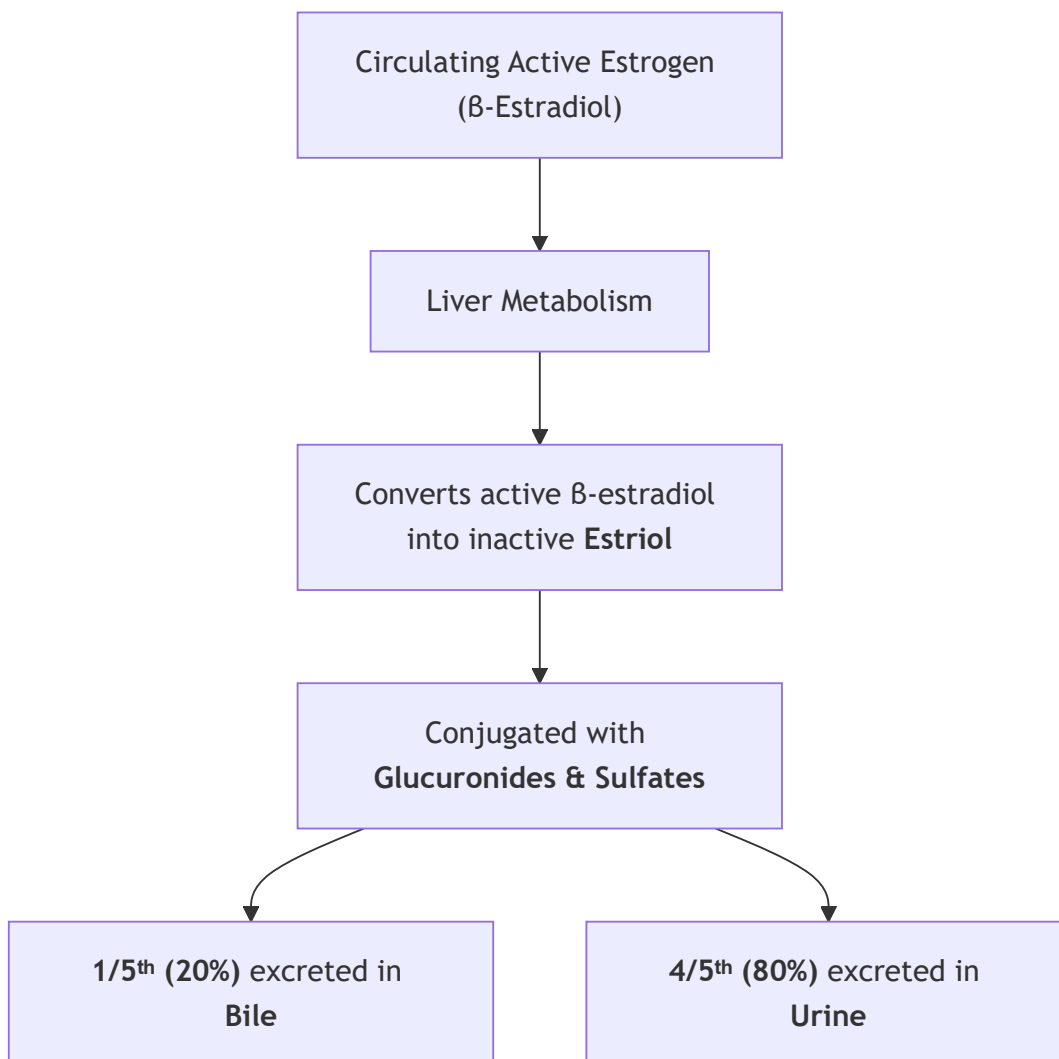
Biosynthetic Pathway

Acetate → Cholesterol → Pregnenolone → Progesterone → Testosterone → **Estrogen**

Transport in Plasma

- Transported bound to **plasma albumin** (primary carrier) and in smaller quantities bound to **sex hormone-binding globulin (SHBG)**.

Metabolism & Excretion



MODE OF ACTION

- **Receptor Location** : Estrogen receptors (ER) are located primarily on the **nuclear membrane / nucleus** of target cells.
- **Receptor Subtypes & Distribution** :—

REGULATION OF ESTROGEN SECRETION

- Regulated predominantly by **Follicle-Stimulating Hormone (FSH)** released from the anterior pituitary.
- **Second Messenger Mechanism** :

FSH binds to membrane receptors on Theca & Granulosa cells

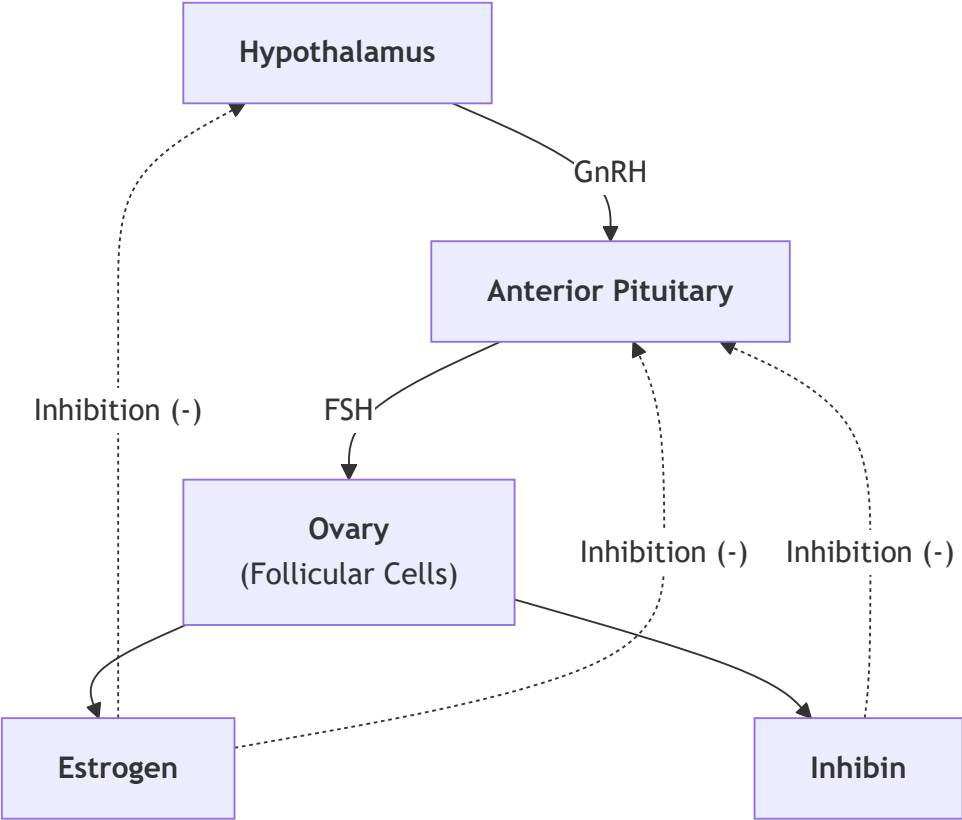


Activates intracellular **cAMP** second messenger system



Stimulates secretory and aromatase enzymatic activities → ↑ **Estrogen Synthesis**

Hypothalamo-Pituitary-Ovarian Feedback Loop



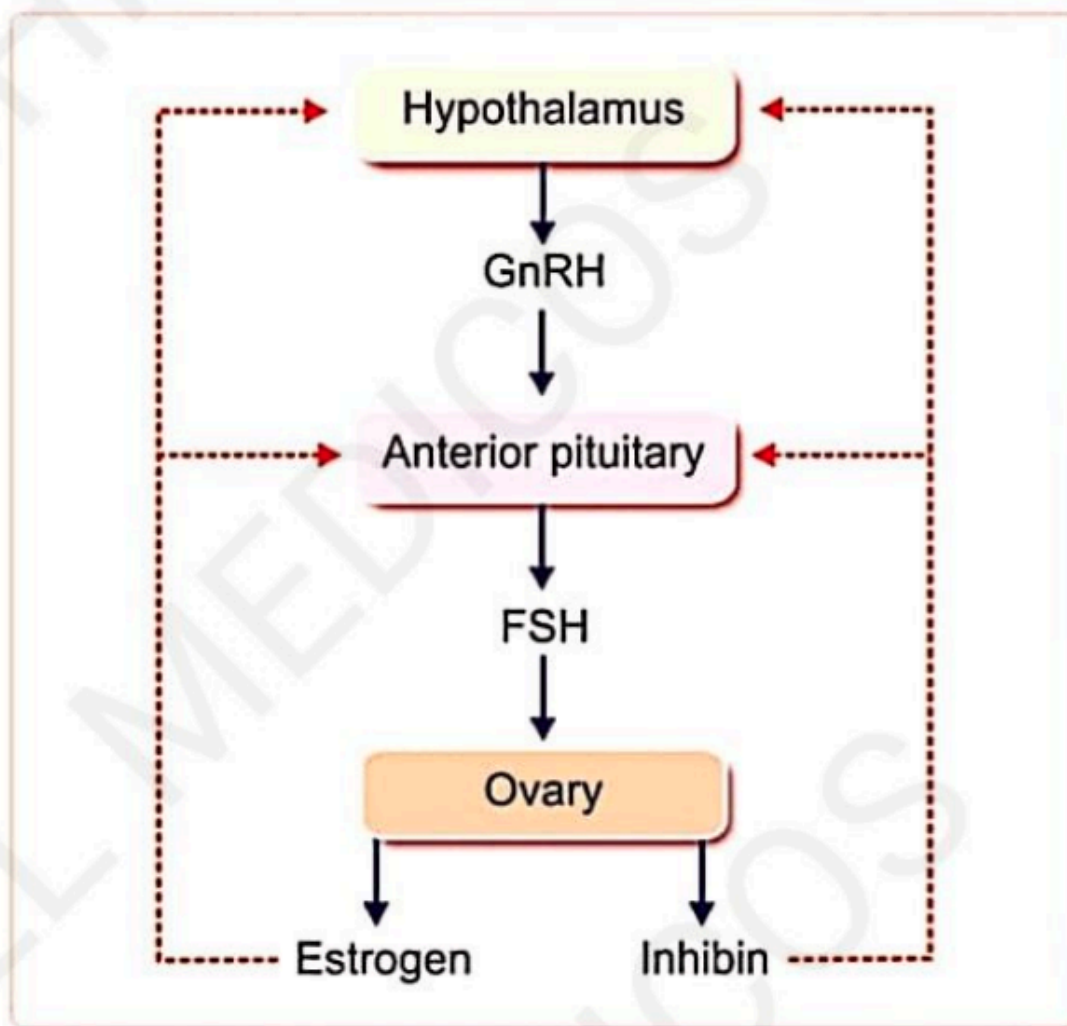


FIGURE 79.2: Regulation of estrogen secretion.
Red dotted lines indicate inhibition.

FUNCTIONS OF ESTROGEN

- **Primary Role :** Promotes **cellular proliferation and tissue growth** in sexual organs and other tissues related to reproduction.

1. ACTIONS ON FEMALE REPRODUCTIVE ORGANS

Target Organ	Physiological Actions & Morphological Changes
1. Ovarian Follicles	• Promotes growth and maturation of ovarian follicles by stimulating the proliferation of follicular (granulosa) cells. • ↑ Increases the secretory activity of theca cells.

Target Organ	Physiological Actions & Morphological Changes
2. Uterus	• Size: Enlarges the uterus to about double its original size via cellular proliferation. • Endometrium: ↑ Increases blood supply; stimulates proliferation and dilation of spiral vessels; promotes deposition of glycogen and lipids. • Myometrium: ↑ Increases spontaneous excitability and contractility of uterine smooth muscle; ↑ increases sensitivity to oxytocin.
3. Fallopian Tubes	• Acts on the mucosal lining → ↑ increases the number and size of epithelial cells (ciliated cells). • ↑ Increases ciliary activity and beat frequency to facilitate the transport of the ovum towards the uterus. • Enhances proliferation of secretory/glandular cells.
4. Vagina	• Transforms vaginal epithelium from delicate cuboidal to stratified squamous epithelium (making it more resistant to trauma and infection). • ↑ Increases epithelial cell layers through rapid proliferation. • ↓ Reduces vaginal pH (increases acidity via lactic acid production from glycogen), protecting against common vaginal infections such as <i>gonorrheal vaginitis</i>.

2. ACTIONS ON BREASTS & SECONDARY SEXUAL CHARACTERISTICS

Effect on Breasts

- Promotes development of stromal connective tissues of the breasts.
- Drives the growth of an **extensive ductile system**.
- Promotes adipose tissue deposition in the ductile framework.
- Stimulates the development of lobules and alveoli to some extent.

Effect on Secondary Sexual Characteristics

Feature / System	Characteristic Changes
Hair Distribution	• Hair develops in the pubic and axillary regions. • Pubic hair pattern has the base of the triangle pointing upwards (horizontal

Feature / System	Characteristic Changes
	upper border). • Scalp hair grows profusely.
Skin	Becomes soft, smooth, and warm (↑ increased cutaneous vascularity).
Body Shape	Shoulders become narrower, hips broaden, thighs converge, and arms diverge (feminine contour).
Pelvis	• Broadening of the pelvis (↑ increased transverse diameter). • Assumes a round / oval shape. • Pelvic outlet becomes round/oval-shaped to facilitate childbearing.
Voice	Retains a characteristic high-pitch voice .

3. SYSTEMIC & METABOLIC ACTIONS

- **Effect on Bones :—**

- ↑ Increases **osteoblastic activity** and bone matrix formation.
- Causes **early fusion of the epiphyses** with the shaft of long bones.
- **Clinical Correlation (Osteoporosis):** In postmenopausal women / old age, estrogen secretion becomes deficient → bones become extremely porous, fragile, and highly susceptible to fractures (**Osteoporosis**).

- **Effect on Metabolism :—**

- **Protein Metabolism:** Exerts a mild protein anabolic effect (↑ increases total body protein).
- **Fat Metabolism:** Stimulates deposition of fat in subcutaneous tissues, especially in **breasts, buttocks, and thighs**.

- **Effect on Electrolyte & Fluid Balance :—**

- Promotes Na^+ and H_2O **retention** by renal tubules, causing mild fluid retention during high estrogen states.